

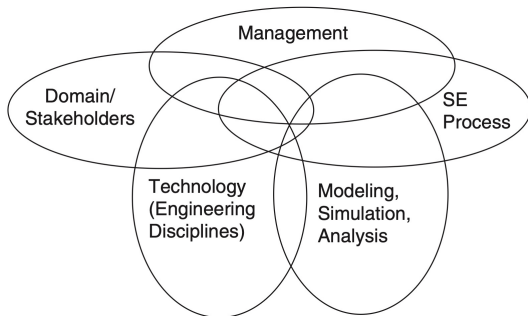
Project Planning Process & Project Charter Components

DAEN 459 / Week 2 / Fall 2025

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Previously ... team formation



Expertise required on the engineering team for a system.

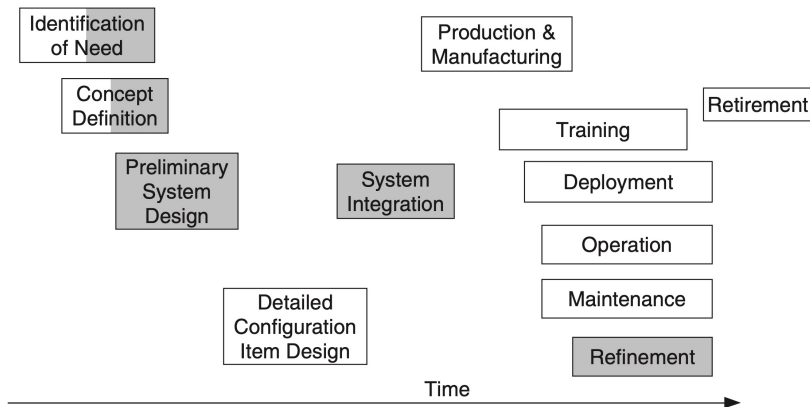
Project Planning & the Project Charter

- ▶ **Gather information:** Consolidate meeting notes and stakeholder needs.
- ▶ **Clarify objectives:** Scope, goals, business case.
- ▶ **Stakeholder alignment:** Roles, responsibilities, approvals.
- ▶ **Project structure:** Deliverables, milestones, timeline overview.
- ▶ **Risk awareness:** Key assumptions, constraints, and potential risks.

Project Planning Process

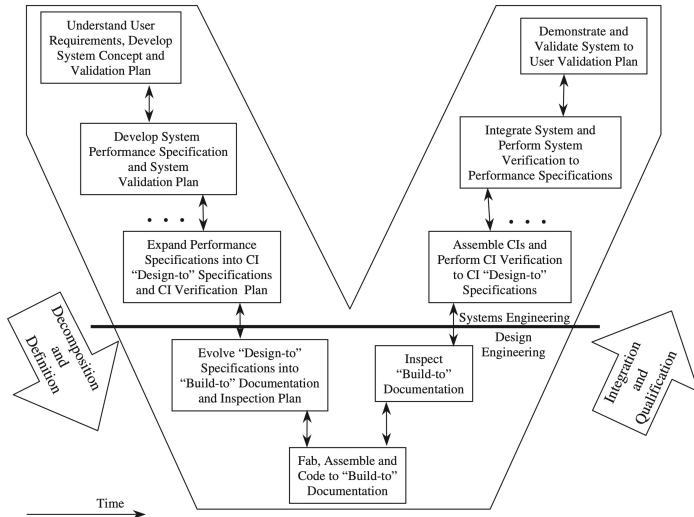
- ▶ Define scope, objectives, and deliverables
- ▶ Identify stakeholders and gather requirements
- ▶ Develop the Work Breakdown Structure (WBS)
(draft, to be refined through iteration)
- ▶ Estimate resources, time, and costs
- ▶ Sequence activities and create a project schedule
- ▶ Identify risks and plan mitigations
- ▶ Finalize project charter (project management plan)

Phases of the System Life Cycle



One of many possible representations of the life cycle of a system.

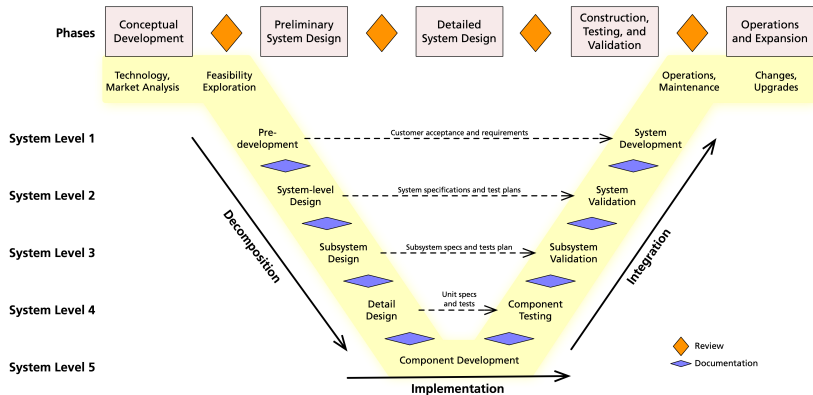
V-model



Systems engineering "Vee" after Forsberg and Mooz [1992].

(Extracted from "The engineering design of systems - models and methods", p.10.)

V-model



Ideally we should aim one level for every 1-2 weeks.
(Prepared by the author, adapted for use in this course.)

Project Charter Components

- ▶ **Purpose & Objectives:** why the project exists
- ▶ **Scope Statement:** boundaries of work
- ▶ **High-Level Requirements:** essential deliverables
- ▶ **Stakeholders & Roles:** sponsor, team, clients
- ▶ **Milestones & Timeline:** major phases and deadlines
- ▶ **Budget Overview:** resource allocation, all kinds (e.g., skills, costs, equipment, and time)
- ▶ **Risks & Assumptions:** known challenges, internal/external threats
- ▶ **Validation & Verification:** how to meet intended purposes
- ▶ **Approval & Sign-Off:** formal authorization

Validation & Verification (V&V)

- ▶ **Validation:** are we building the *right* solution?
Ensure alignment with sponsor needs and project objectives.
Review requirements and constraints early.
- ▶ **Verification:** are we building the solution *right*?
Test deliverables against defined requirements.
Iterative reviews, simulations, and peer evaluations.
- ▶ Continuous process throughout **project lifecycle**.

Goal for This Week: Find an Advisor

- ▶ **Primary advisor (required):** ISEN faculty member.
(I can recommend matches once I review your initial notes)
- ▶ **Role:** subject-matter expert; offers guidance, feedback, and technical direction.
- ▶ **Cadence:** meet about every two weeks unless an alternative schedule is approved.
- ▶ **Professionalism:** treat advisors as part of the team: communicate clearly and respectfully. (ensure a formal, businesslike approach at all times).
- ▶ **Secondary advisor?** (optional): from TAMU or even external (needs approval with brief justification).

Which should we focus on?

